# Goutam Barman

**Machine Learning | Data Analytics | Full-Stack Development**

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GitHub: https://github.com/ipgoutam | Portfolio: Goutam Barman | Computer Science Portfolio

## PROFESSIONAL SUMMARY

Aspiring Machine Learning and Full-Stack Developer with hands-on experience building machine learning models, data analytics dashboards, and full-stack web applications. Proficient in Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, MERN Stack, Tableau, and Power BI. Experienced in data preprocessing, exploratory data analysis, predictive modeling, data visualization, REST APIs, database integration, and responsive web development.

## TECHNICAL SKILLS

**Programming:** Python, Java, JavaScript, HTML, CSS

**Machine Learning:** Scikit-learn, Supervised Learning, Classification, Regression, Predictive Modeling, Data Preprocessing, Feature Engineering, Model Evaluation, EDA

**Python Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

**Data Analytics:** Power BI, Data Cleaning, Data Visualization, Dashboard Development, Business Intelligence, KPI Analysis

**Web Development:** MongoDB, Express.js, React.js, Node.js, MERN Stack, REST API, Full-Stack Development

**Tools:** Git, GitHub, Jupyter Notebook, VS Code

## PROJECTS

**CineMatch – AI Movie Recommendation Platform** | Python, Scikit-learn, Pandas, FastAPI, REST APIs, TF-IDF, Cosine Similarity, HTML, CSS, JavaScript, TMDB API

- Developed a full-stack movie recommendation system using content-based filtering, TF-IDF vectorization, and cosine similarity to generate movie recommendations based on content similarity.
- Implemented machine learning workflows and text feature extraction** to process movie metadata and identify similarities between movies.
- Built RESTful APIs using FastAPI to integrate the recommendation engine with the frontend and serve movie search and recommendation results.
- Integrated the TMDB API to retrieve movie metadata, descriptions, posters, and other information for enhanced movie discovery.
- Developed a responsive frontend with HTML, CSS, and JavaScript, featuring movie search, recommendations, movie details, and watchlist functionality.
- Deployed the application on Render, integrating the frontend, backend, and machine learning components into a live web platform.

**Live Demo:** https://cinematch-1-wpgm.onrender.com/

**WonderLust — Hotel & Tourist Booking Platform** | *MongoDB, Express.js, React.js, Node.js, JavaScript*

- Developed a full-stack hotel and tourist accommodation booking platform using the MERN Stack.
- Built RESTful APIs using Node.js and Express.js for application and booking operations.
- Integrated MongoDB for storing and managing users, hotels, listings, and booking-related data.
- Developed a responsive frontend using React.js, JavaScript, HTML, and CSS.
- Implemented frontend-backend communication through REST APIs for a seamless booking workflow.

**Global Airbnb Performance Dashboard | Power BI**

- Developed an interactive Power BI dashboard to analyze Airbnb's global market performance across 10 major cities.
- Performed data cleaning, data transformation, and preprocessing to improve data quality and prepare datasets for analysis.
- Created DAX measures and calculated KPIs to track listing performance, host distribution, property types, and market trends.

- Analyzed listing growth, city-wise market share, and property categories, including entire places, hotel rooms, private rooms, and shared rooms.
- Compared average prices across property types to identify pricing differences and understand accommodation market trends.
- Evaluated customer ratings across accuracy, cleanliness, communication, location, and value for money to compare customer satisfaction across cities.
- Designed interactive KPI cards, charts, trend lines, and visualizations to present key insights and support data-driven decision-making.

**Live Demo:** https://app.powerbi.com/links/fXlwLDJ9wK?ctid=a22bf06c-9316-436a-a631-65a726549c1e&pbi_source=linkShare

## EDUCATION

**Bachelor of Technology (B.Tech) — Artificial Intelligence & Machine Learning**
[Siliguri institute of technology] | [Siliguri, West Bengal] | Expected Graduation: [2028]
**Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Python Programming, Statistics, Data Analytics, Web Development

## CERTIFICATIONS

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## ADDITIONAL INFORMATION

**Languages:** [English, Bengali, Hindi]
**Interests:** Machine Learning, Data Analytics, Artificial Intelligence, Full-Stack Development